JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 98, NO. A8, PAGES 13,409–13,422, AUGUST 1, 1993

Structure of the Dayside Magnetopause for Low Magnetic Shear

G. PASCHMANN, W. BAUMJOHANN, N. SCKOPKE, AND T.-D. PHAN

Max-Planck-Institut für extraterrestrische Physik, Garching, Germany

H. LÜHR

Institut für Geophysik und Meteorologie, Braunschweig, Germany

We have analyzed 22 AMPTE/IRM satellite passes through the low-latitude magnetopause region for which the magnetic shear, i.e., the field rotation angle on transit from the magnetosheath to the magnetosphere, was less than 30°. We found that on all passes a key time could be identified where the proton temperature, and usually also in the electron temperature and the temperature anisotropies, show net and often discontinuous changes. A change in plasma flow direction also occurs at this time. The small field rotations that occur have no fixed relationship to the key time. Earthward of the key time, the plasma has the characteristics of the magnetopause boundary layer, i.e., reduced density and bulk velocity, and in particular an electron temperature anisotropy with $T_{e\parallel} > T_{e\perp}$, until further and usually more dramatic changes in plasma thermal properties mark the entry into the magnetosphere proper. Boundary layer durations varied widely, from 4 s to 14 min. On the magnetosheath side we observed a layer of plasma density depletion (by a factor of two or more) and concurrent magnetic field pile-up in less than half the crossings. This depletion layer was 3 min wide on average and often characterized by a drop in $T_{p\parallel}$, as predicted. From the observational evidence we conclude that the changes in plasma thermal and flow properties at the key time mark the crossing of the magnetopause under conditions of low magnetic shear. It is tempting to attribute these changes to the crossing of a topological boundary, e.g., a transition from open interplanetary to closed geomagnetic field lines. There is evidence, however, that in some cases the field lines immediately earthward of the key time are not closed. The consistent presence of a boundary layer inside the low-shear magnetopause confirms earlier inferences that solar wind plasma can enter the magnetosphere regardless of field orientation. Inspection of the velocity distribution functions indicates that the plasma is heated upon entry. Taking measured plasma velocities along the magnetopause normal direction, we have inferred an average magnetopause speed of 11 km/s, an indication that the low-shear magnetopause moves more slowly. The duration of the temperature transition translates into a thickness of often less than 50 km, comparable with the typical proton gyroradius. For the plasma depletion layer and the boundary layer we obtained average thicknesses of 0.4 and 0.3 R_E, respectively.

1. INTRODUCTION

The properties of the Earth's magnetopause under conditions of large magnetic shear have been studied extensively since the first in situ measurements in 1961. The identification of the magnetopause is usually straightforward in these cases because it can be based on the large rotation, i.e., large shear, displayed by the magnetic field as the observing spacecraft passes from one side of the magnetopause to the other. Large shear at the subsolar magnetopause implies southward directed interplanetary magnetic field (IMF). For northward IMF, on the other hand, it can happen that the fields on the two sides of the dayside magnetopause become (nearly) parallel to each other, i.e., exhibit low shear. Under these conditions the identification of the magnetopause from magnetometer data alone becomes difficult or impossible, particularly if one notes that the field magnitude often changes little as well. Analysis of magnetopause crossings with large magnetic shear has to a large extent focussed on the search for evidence of magnetic reconnection across the magnetopause, a process favored by large field rotation angles. Low-shear cases, on the other hand, are of interest because one expects the signatures of reconnection to be absent. Therefore other transport processes, such as diffusion, impulsive penetration, etc. should become more evident. To date, the case of low shear has not received the same attention, a major reason certainly being the noted difficulty with the identification of the magnetopause.

One of the few low-shear dayside magnetopauses studied previously is the November 5, 1977, crossing by the ISEE 1 and 2 satellites. It belongs to a set of magnetopause crossings which served as a reference for much of the early ISEE work, and is characterized by a field rotation across the magnetopause of only 20° and very little change in field magnitude [*Russell and Elphic*, 1978; *Paschmann et al.*, 1978]. A low-shear crossing, observed in the low time resolution data from Heos 2, was reported by *Haerendel et al.* [1978]. Recently, *Song et al.* [1990*a*] have extensively analyzed a case observed by ISEE where the field rotated very little and the transition from magnetosheath to magnetosphere showed a complicated structure.

With the goal of studying the nature of the magnetopause and the transport processes when low-latitude reconnection should not be important, we have surveyed the AMPTE/IRM data and obtained a set of 22 low-latitude dayside passes (cf. Figure 1) where the classification of the magnetopause as low-shear was un-

Copyright 1993 by the American Geophysical Union.

Paper number 93JA00646.
0148–0227/93/93JA–00646$05.00

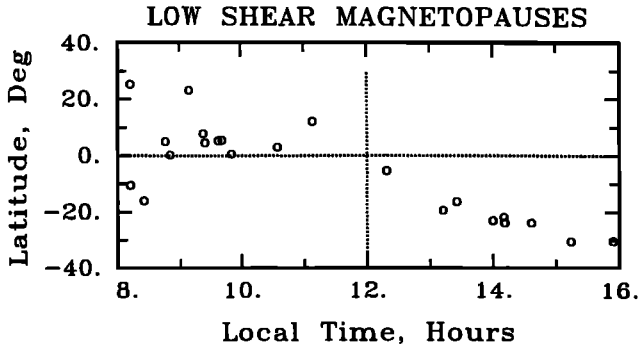

Fig. 1. Location of the 22 AMPTE/IRM low-shear magnetopause crossings in GSM latitude and local time.

ambiguous. The present paper will discuss signatures which appear to mark the magnetopause under low-shear conditions, and their implications for the structure of the boundary region.

The data used in this study were obtained with the three-dimensional plasma instrument and the three-axes flux-gate magnetometer aboard the AMPTE/IRM satellite [*Paschmann et al.*, 1985 *Lühr et al.*, 1985]. The plasma instrument measures the full 3D distribution functions of ions and electrons in a single spacecraft revolution (about 4.4 s), in the range from 20 eV to 40 keV, and 15 eV to 30 keV, respectively. Moments of the distributions were routinely calculated on board and transmitted at the full time resolution. The actual 3D distribution functions were transmitted with reduced velocity space resolution and less frequently, typically every other spin. In a few cases the onboard moment computation failed and the moment data were computed on the ground from the transmitted distribution functions. The magnetometer data, measured with 32 samples/s, are presented as averages over the plasma sampling times.

2. Survey of Low-Shear Crossings

A survey of the AMPTE/IRM dayside data revealed a number of passes where according to the measured plasma density profiles a transition from the magnetosheath to the magnetosphere must have occurred at some time, but the magnetic field changed very little. In the following we present six examples to illustrate the range of conditions observed.

2.1. *Sample Cases*

September 17, 1984. Figure 2 shows plasma and magnetic field data for an inbound pass on September 17, 1984, at about 1400 hours local time and −22° GSM latitude. At the start, the spacecraft is located in the magnetosheath, as evidenced by the high plasma density, low temperatures, and fluctuating magnetic field strength and direction. At 10:50:20 UT, the low plasma density, high temperature, and stable magnetic field all ascertain that the spacecraft has entered the magnetosphere. The field angle φ_B has become close to 0° because in the magnetosphere the field is virtually aligned with the L axis from which φ_B is measured.

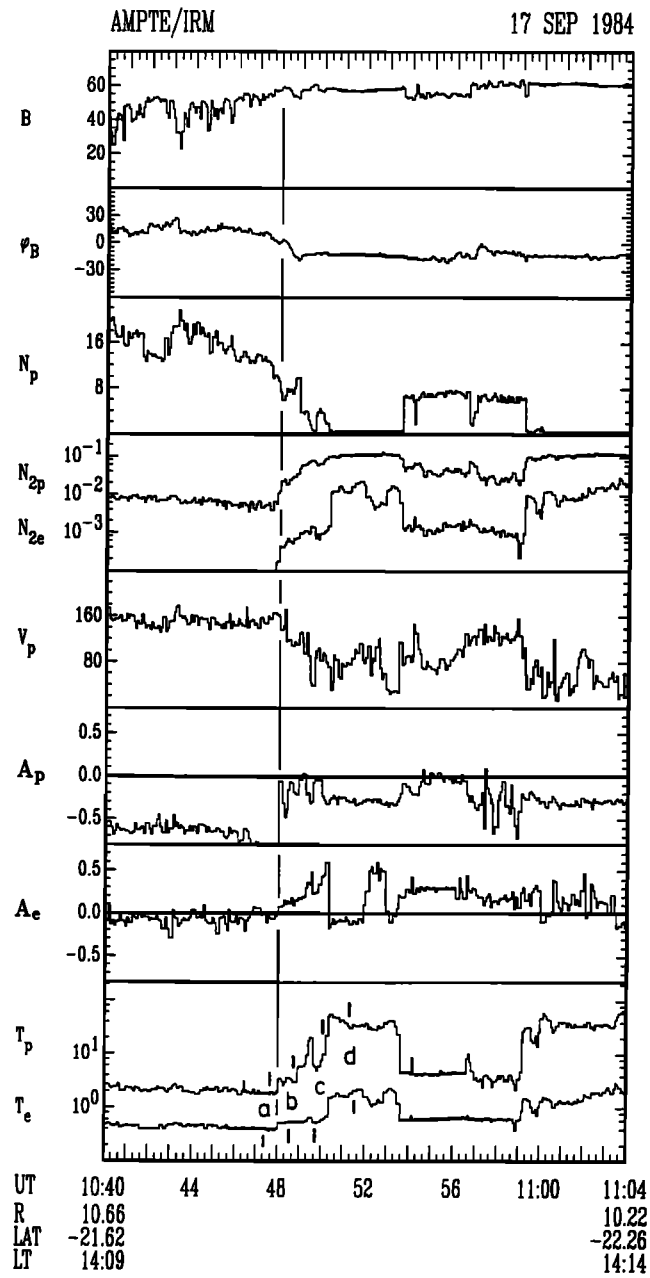

Fig. 2. Inbound crossing of the magnetopause on September 17, 1984. From top to bottom, the figure displays the magnetic field strength B (in nT); the angle φ_B the field makes with the L axis in the L,M plane of the LMN boundary normal coordinate system where the normal is taken from a standard magnetopause model; the proton number density N_p (in cm^{-3}); the partial densities N_{2p} and N_{2e} of energetic ions and electrons in the energy range 8 keV to 40 keV and 1.8 keV to 30 keV, respectively; the proton bulk speed V_p (in km/s); the proton and electron temperature anisotropies A_p and A_e, defined as $T_\parallel/T_\perp - 1$; and the proton and electron temperatures T_p, T_e (in 10^6K). Data points are spaced 4.4 s apart. Universal time and spacecraft location are given across the bottom, as well as the magnetopause normal direction, taken from a model magnetopause surface. The short vertical bars, denoted $a-d$, in the temperature panel mark the times of the distribution functions presented in Figures 11 and 12. Note the simultaneous increase in the temperatures and temperature anisotropies at 10:48:02 UT, marked by the vertical line. At 10:50:20 UT the magnetosphere proper was entered, as evidenced by yet another temperature increase.

21562202a, 1993, A8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/93JA00646 by University College London UCL Library Services, Wiley Online Library on [10/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

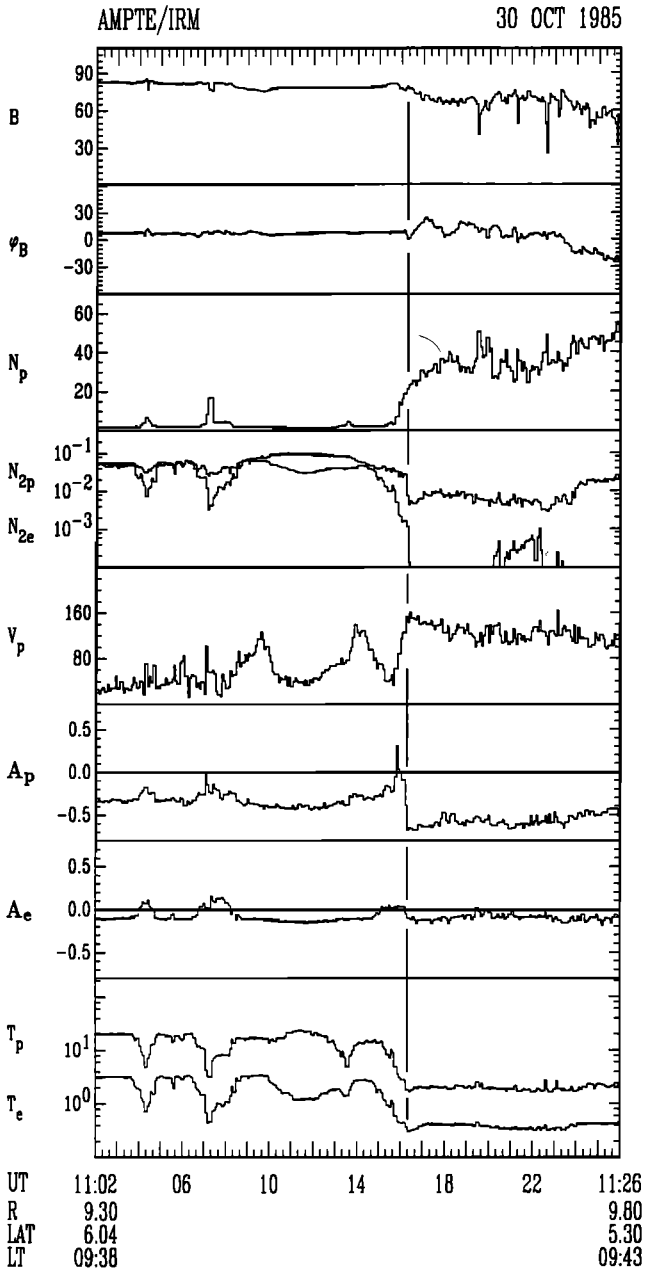

Fig. 3. Outbound pass through the magnetopause region on October 30, 1985. Sudden changes in temperatures and anisotropies occur at 11:16:13 UT. The boundary layer encounter which precedes this time is less than about 0.5 min long. Again there is evidence for plasma depletion and magnetic field pile-up.

Prior to entry into the magnetosphere proper, at 10:48 UT, there is a very sudden rise in the electron and proton temperatures, and in their anisotropies, in particular that of the protons. The magnetic field direction and magnitude, on the other hand, showed only a small and slow variation at this time. As we will see, such changes in the plasma thermal properties are symptomatic for all low-shear crossings we have observed. We suggest that they are signatures of the crossing of the low-shear magnetopause.

Taking 10:48 UT as the magnetopause crossing, one would conclude from the density profile that a boundary layer (i.e., a layer of magnetosheath-like plasma Earthward of the magnetopause) was observed until 10:50:20 UT, at which time the magnetosphere proper was entered as already noted earlier.

The gradual decrease in density and simultaneous increase in magnetic field strength, observed prior to 10:48 UT, then becomes an exterior feature, a field pile-up and plasma depletion layer sunward of the magnetopause. Note that the proton temperature anisotropy parameter, A_p, becomes substantially more negative in this layer. As we define the anisotropy parameter A as $T_{\|}/T_{\perp} - 1$, this means the ratio $T_{p\|}/T_{p\perp}$ is decreasing.

It should be noted that another encounter with a magnetosheath-like plasma started at 10:55:30 UT. By a comparison with the density levels registered earlier it appears probable that the satellite is again encountering the boundary layer which happens to be isolated from the previous encounter, either temporally or spatially. However, here one is faced with a dilemma. The boundary layer is located on terrestrial field lines, and therefore the field orientation will not vary much on entry into or exit from it. But this is, by definition, also true when crossing a low-shear magnetopause. Therefore a distinction between multiple low-shear magnetopause crossings and encounters of an isolated boundary layer is not possible a priori. Some independent signature of the boundary layer is required to make that distinction. As we will see, the boundary layer is usually characterized by positive electron temperature anisotropies, while in the magnetosheath the anisotropy is usually slightly less or equal to zero. Note again that the anisotropy parameter has been defined as $T_{\|}/T_{\perp} - 1$. Thus $T_{e\|} > T_{e\perp}$ in the boundary layer. Using this criterion, the interval after 10:55:30 UT would indeed be classified as the boundary layer, and not as a pair of low-shear crossings of the magnetopause.

October 30, 1985. This example, shown in Figure 3, is from an outbound pass, and thus the sense of the changes in temperatures and anisotropies, observed at 11:16:13 UT, is reversed. There is little change in the field direction at that time. The boundary layer encounter is shorter in this case, less than about 0.5 min, and, as appropriate for an outbound pass, occurs prior to the "magnetopause" crossing. Note again the gradual changes of density and magnetic field strength outside the magnetopause, this time with only a slightly more negative proton temperature anisotropy parameter.

August 30, 1985. Figure 4 shows an outbound crossing on August 30, 1985, where marked changes in the plasma thermal properties occur at 15:15:00 UT. This time, however, the electron temperature anisotropy changes only little. Furthermore, when looking at the time series from the magnetosheath side, there is a "precursor" (at 15:17:00 UT) which seems to be a first approach to the magnetopause, but no actual crossing. Inspection of the magnetic field angle shows that at either time the field is extremely steady. Several minutes later, at 15:21:30 UT the field rotates by about 30°. No change in plasma thermal properties occurs at this time. We believe that the field rotation marks the passage of a directional discontinuity in the IMF. There are

21562202a, 1993, A8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/93JA00646 by University College London UCL Library Services, Wiley Online Library on [10/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

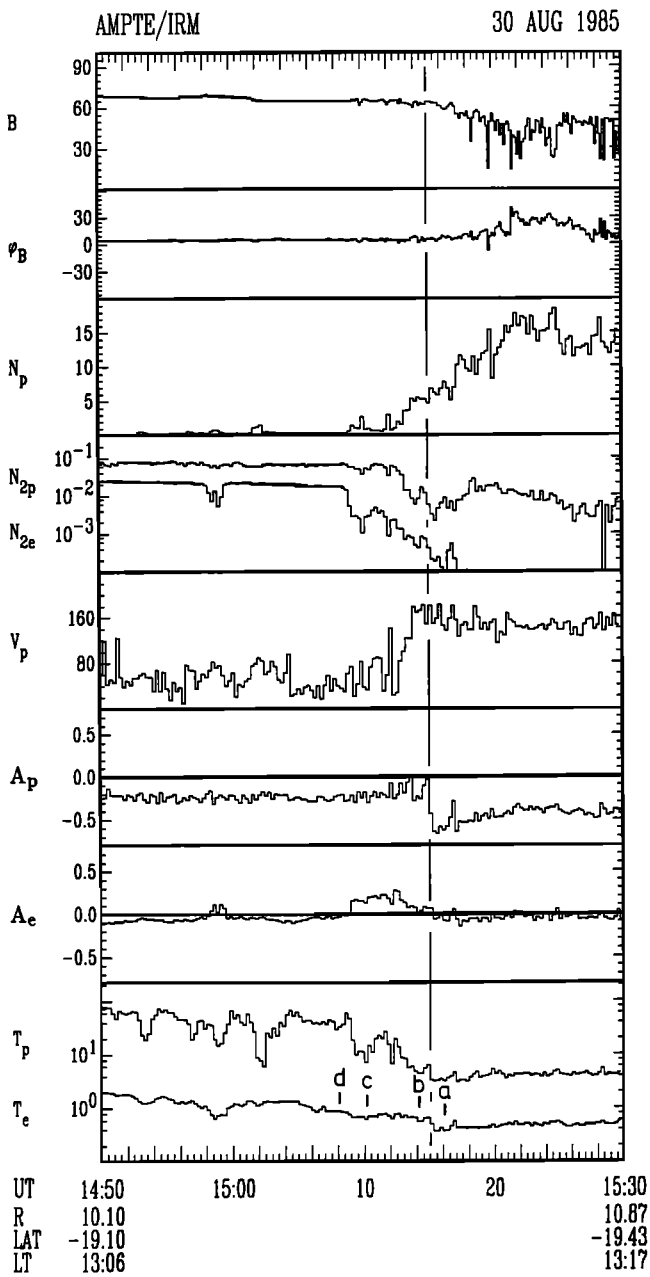

Fig. 4. Outbound crossing on August 30, 1985, with the primary thermal changes at 15:15:00 UT, but a precursor at 15:17:00 UT. Note the extraordinary steady magnetic field angle at these times. Plasma data points are averaged over 15 s because the proton moments had to be computed from the transmitted distribution functions in this case. The short vertical bars denoted $a-d$ mark the times of the electron distributions shown in Figure 13.

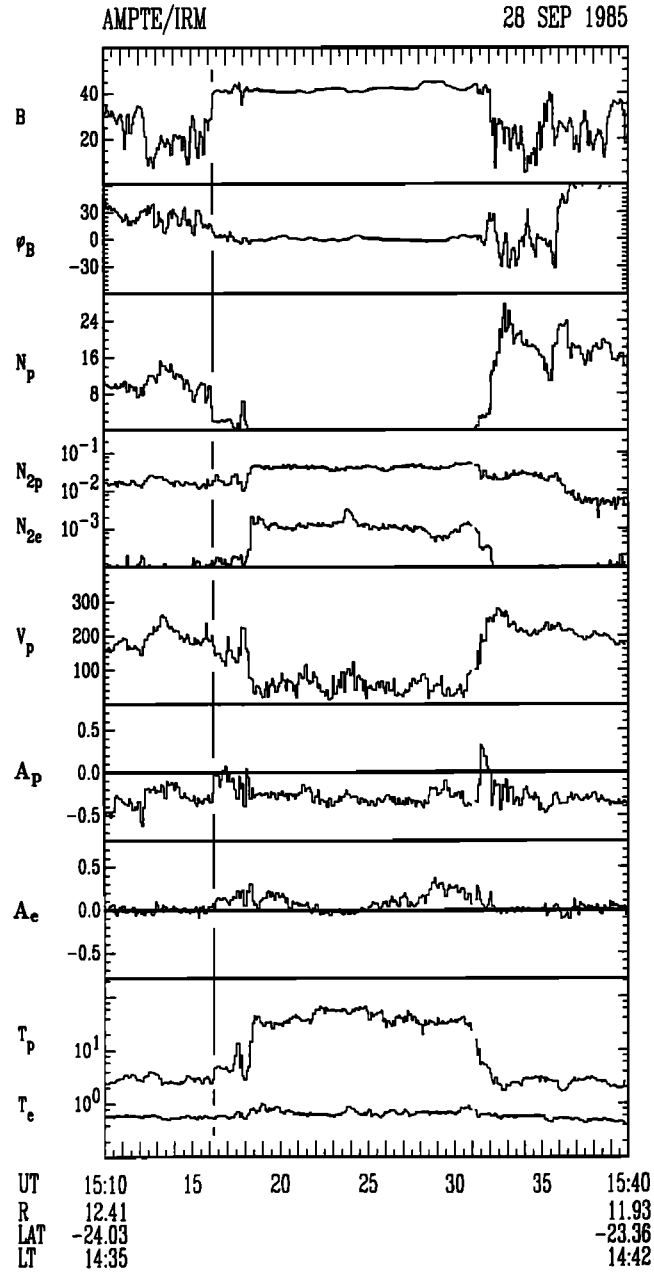

Fig. 5. A pass on September 28, 1985, with a low-shear magnetopause crossing at 15:16:16 UT and another crossing at 15:30:00 UT. Note that, contrary to the previous examples, the plasma density is enhanced and the magnetic field strength depleted on the sheath side.

no IMP 8 data for this time, so this hypothesis cannot be tested.

Regardless of which time one takes for the magnetopause, 15:15 UT or 15:17 UT, it is clear from Figure 4 that there is a significant density depletion on the magnetosheath side, accompanied by a more negative proton temperature anisotropy parameter.

Note that the boundary layer in Figure 4 again is characterized by a positive electron temperature anisotropy. At the inner edge (at 15:09 UT), the ion temperature rises sharply, but the energetic ion density stays nearly constant. The electrons show the opposite behavior in this case: their overall temperature stays nearly constant while the energetic component rises sharply. This point will be taken up in Section 4.

September 28, 1985. The density depletion outside the magnetopause noted in the previous examples is not a universal feature. Figure 5 shows a pair of crossings on September 28, 1985, where there is actually a maximum in density, accompanied by a minimum in magnetic field strength, outside the magnetopause which lasts for several minutes. The same feature is also observed on another crossing (November 17, 1985) in our data set. Figure 5 also illustrates that the electron temperature is not always a reliable magnetopause indicator. In neither crossing does it show a notice-

21562202a, 1993, A8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/93JA00646 by University College London UCL Library Services, Wiley Online Library on [10/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

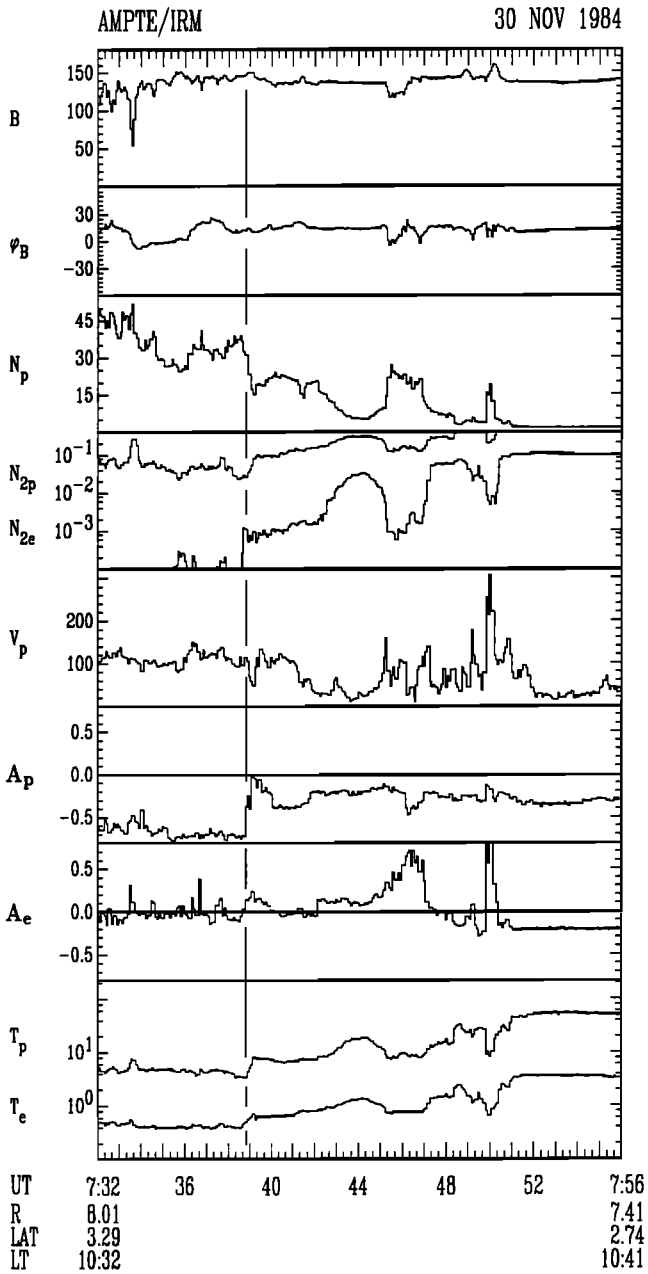

Fig. 6. Inbound pass on November 30, 1984, demonstrating that the temperature and anisotropy transitions need not be as sudden as in the earlier examples. The temperature changes occur at 07:39 UT, followed by what is interpreted as a boundary layer until about 07:42:30 UT. The boundary layer encounter after 07:45 UT could also be interpreted as two more magnetopause crossings. The positive electron temperature anisotropy, however, distinguishes it from the true magnetosheath where the anisotropy is close to zero. Note the appearance of a peak in magnetic field strength at the key time.

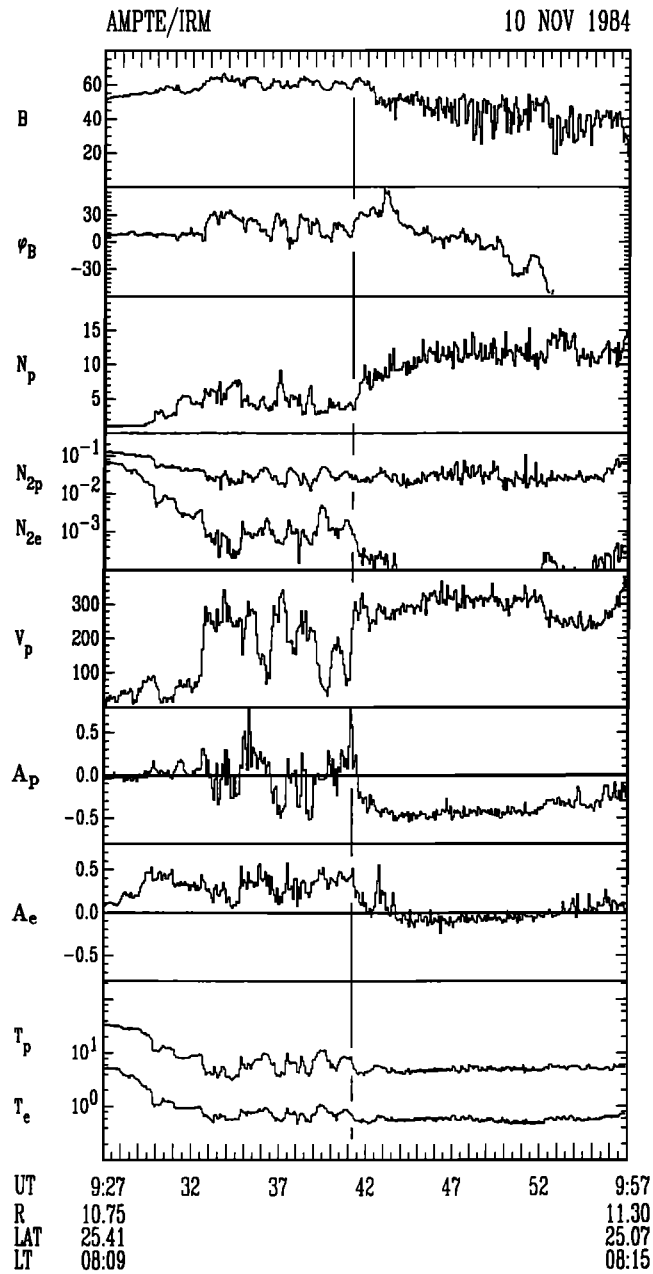

Fig. 7. Inbound pass on November 10, 1984, for which we place the magnetopause at around 09:41:20 UT, when the ion and electron temperatures and their anisotropies change. *Hall et al.* [1990, 1991], using AMPTE/UKS data, have placed the magnetopause at about 09:52 UT where the field shows a substantial rotation. Note the broad boundary layer, starting at 09:29:50 UT, distinguished from the magnetosheath by the positive electron temperature anisotropy.

able change. The electron temperature anisotropy, on the other hand, again rises to a positive value in the boundary layer.

November 30, 1984. The pass on November 30, 1984, shown in Figure 6, serves to illustrate that the change is plasma thermal properties are not always as sudden as in the previous examples. Here they are spread over almost 30 s, around 07:39 UT. (In another crossing (on October 13, 1985, not shown), the transition even takes 3 min). A boundary layer is observed until about 07:42:30 UT, followed by what is probably another boundary layer encounter at 07:45 UT. Note the characteristic positive electron temperature anisotropy parameter in both intervals which distinguishes them from the magnetosheath. There is clear evidence for a peak in field strength right at the time of the thermal changes. No plasma depletion layer is apparent in this crossing.

November 10, 1984. As a final example we consider the crossing on November 10, 1984 (Figure 7). Based on the changes in temperatures and their anisotropies, we place the magnetopause at about 09:41:20 UT, although

21562202a, 1993, A8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/93JA00646 by University College London UCL Library Services, Wiley Online Library on [10/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

these changes are not as pronounced as in previous examples. A plasma depletion layer is again evident, but this time the proton temperature anisotropy parameter actually increases. The boundary layer is unusually extended, and is characterized by a field strength significantly higher than on either side.

The November 10, 1984, crossing has been discussed extensively in a paper by *Hall et al.* [1990] based on AMPTE/UKS electron and magnetic field data. As IRM and UKS on that date are separated by only a few tens of km, the two data sets should be directly comparable. *Hall et al.* find evidence for open field lines in the boundary layer for this crossing. The evidence consists in the observation of unbalanced fluxes of 100–400 eV electrons parallel and anti-parallel to the magnetic field. In another paper [*Paschmann et al.*, 1990] which analyzes in detail the December 29, 1984, low-shear crossing, we had mentioned that the November 10 crossing shows evidence of a finite, although possibly small normal component of the magnetic field.

In spite of these similarities, there is disagreement on the time of the magnetopause crossing. Hall et al. place the magnetopause at 09:52 UT, at which time the magnetic field turns southward. As Figure 7 shows, no changes in electron and ion temperatures or anisotropies are observed at this time. In our opinion this time marks the passage of a discontinuity in the solar wind rather than the magnetopause. As no simultaneous measurements of the IMF appear to be available, this hypothesis cannot be tested. Note that the conclusion on open boundary layer field lines referred to earlier is not affected by the uncertainty of the magnetopause identification, because the open field region identified by Hall et al. extends Earthward of the magnetopause regardless of which time one chooses.

2.2. *Survey Results*

The survey of all AMPTE/IRM crossings in the 08–16 GSM local time range has produced a total of 22 passes where a magnetopause with less than 30° shear could be identified. The location of the crossings is shown in Figure 1. It is likely that some passes involved multiple low-shear crossings. But for reasons discussed earlier, only the outermost transition unquestionably marks a magnetopause crossing. Any other low-shear magnetopause transitions further Earthward on the same pass could not a priori be distinguished from passes into and out of the boundary layer.

The situation becomes less clear if in a series of multiple magnetopause crossings the outermost ones are high-shear crossings, which are followed by crossings which have become low-shear as a result of a temporal change in the direction of the IMF. Here an unambiguous classification of a transition as a low-shear magnetopause (rather than a boundary layer encounter) is only possible if one can be certain the medium on one side of the transition really is the magnetosheath. That often can be determined rather reliably from the fluctuation levels in the magnetic field. In other cases counting the number of crossings can uniquely clarify the situation if one notes that there must be an odd total number of magnetopause crossings in a given pass, or that after any even number of crossings one has returned to the medium one started out in. We have only included a low-shear crossing occurring on the same pass with high-shear crossings if one or the other method succeeded.

We could have used the electron temperature anisotropy to help distinguish low-shear magnetopause from boundary layer crossings, but did not do so because we did not want the results to depend on the validity of the anisotropy criterion.

The results of the survey of the 22 passes included in this study are summarized in Table 1. The table lists the times where distinct net changes in the plasma thermal properties were observed, and also identifies (by the plus and minus signs) which of the temperatures or anisotropies participated in the change, and which did not. Multiple pluses imply changes in the associated quantities occurring within no more than 10 s of each other. No case was observed where no such change in thermal properties was observed. For case 14 no electron data exist and thus no statement on the electron behavior can be made.

Table 1 shows that the ion temperature changed in all 22 cases. Of the 21 cases where all four quantities were measured, there are 16 (76%) where three of the four quantities changed simultaneously, and 15 (71%) where both temperatures were involved. All 4 quantities changed simultaneously in 8 (38%) of the cases, and there was one case (20) where only the proton temperature changed. It thus appears that the plasma thermal properties constitute a reliable signature of the magnetopause transition under conditions of low magnetic shear.

TABLE 1. AMPTE/IRM Low-Shear Magnetopause Crossings

	Date	UT	R	LT	λ	i/o	T_e	T_p	A_e	A_p
1	Sep. 17, 1984	10:48:02	10.5	14.2	-21.8	i	+	+	+	+
2	Sep. 21, 1984	13:10:00	9.8	11.1	12.1	o	+	+	+	+
3	Oct. 28, 1984	12:01:39	11.3	9.2	23.2	o	+	+	+	+
4	Nov. 10, 1984	09:41:20	11.0	8.2	25.3	o	+	+	−	−
5	Nov. 30, 1984	07:38:51	7.8	10.6	3.2	i	+	+	+	+
6	Dec. 9, 1984	10:38:01	10.9	9.4	7.9	i	−	+	+	+
7	Dec. 11, 1984	08:25:34	9.0	9.6	5.4	i	−	+	+	−
8	Dec. 22, 1984	09:27:15	9.7	8.8	5.1	i	+	+	+	−
9	Dec. 27, 1984	21:56:04	10.1	8.4	-16.0	i	+	+	−	−
10	Dec. 29, 1984	18:04:54	10.2	8.2	-10.5	i	+	+	+	−
11	Aug. 19, 1985	15:09:30	12.2	14.2	-23.9	o	−	+	+	+
12	Aug. 30, 1985	15:15:00	10.6	13.2	-19.4	o	+	+	+	+
13	Sep. 4, 1985	15:08:45	12.9	16.1	-30.3	i	+	+	+	+
14	Sep. 17, 1985	13:46:12	12.2	15.2	-30.5	i	0	+	0	+
15	Sep. 28, 1985	15:16:16	12.3	14.6	-23.9	i	−	+	+	+
16	Oct. 13, 1985	11:32:07	10.0	14.0	-23.0	i	+	+	+	+
17	Oct. 24, 1985	13:02:47	10.1	13.4	-16.3	i	+	+	+	−
18	Oct. 30, 1985	11:16:13	9.6	9.7	5.6	o	+	+	+	+
19	Nov. 1, 1985	06:58:50	8.9	9.4	4.6	o	+	+	+	+
20	Nov. 3, 1985	05:46:47	11.9	9.8	0.6	o	−	+	−	−
21	Nov. 17, 1985	13:05:56	9.4	12.3	-5.3	i	+	+	−	+
22	Jan. 4, 1986	09:07:54	12.5	8.8	0.3	i	−	+	−	+

Columns 2–6 list date, Universal Time, and location of the 22 crossings in terms of geocentric distance R, GSM local time LT, and latitude λ, in Earth radii, hours, and degrees, respectively. Column 7 shows the sense of each crossing (inbound or outbound). The symbols on the right indicate which of the temperatures T_e and T_p and temperature anisotropies A_e and A_p changed (+) or did not change (−) at the magnetopause crossing time. No electron data were available for case 14 (0).

21562202a, 1993, A8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/93JA00646 by University College London UCL Library Services, Wiley Online Library on [10/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

3. SUPERPOSED EPOCH ANALYSIS

In the previous section we have established that the low-shear magnetopause crossings are associated with sudden net changes in plasma thermal properties. We also found certain systematic variations in other plasma and magnetic field quantities which followed or preceded the thermal changes in a number of cases. To study these relationships more systematically, we have performed a superposed epoch analysis of the crossings, using the times of the proton temperature jump as the key time. For ease of reference, we will refer to the key times as the magnetopause crossing. We have excluded from the analysis cases 14 and 20. The former because no electron data exist, the latter because the proton temperature was the only quantity which showed a change.

We used printouts of the data to precisely identify the times (Table 1). Three minutes of data were included on either side of the key time, unless other magnetopause or boundary layer crossings occurred within the 3 min. In those cases shorter intervals were chosen. For outbound crossings, the time order of the data was reversed. As the absolute magnitude of some quantities varies considerably from one case to the next, we normalized their averages over the entire time interval to 1. The normalized quantities for the 20 cases were then added together logarithmically, using 10-s time bins. Logarithmic averaging reduces the dominance of cases with larger than average dynamic range. With 4.4 s spacing of the measurements, between two and three data points are binned together. For each bin, the error of the mean was also computed.

For the temperature anisotropies the absolute value matters, and normalization is thus ruled out. As signed quantities they also had to be summed linearly. This does not cause problems because the anisotropies have little dynamic range.

The magnetic field direction in the plane tangent to to the (model) magnetopause, φ_B, was also treated differently. As the L axis of the boundary normal coordinate system is close to the field direction in the subsolar magnetosphere, φ_B is automatically small there, but not exactly zero. We therefore normalized the data such that for each case the average on the magnetosphere side of the key time is precisely zero. As we are interested only in the magnitude of the field rotation, we changed the sign of the angle on the magnetosheath side whenever its average there was negative. The 20 cases were then averaged linearly.

The plasma flow direction φ_V, which we included in the superposed epoch analysis even though it was not shown in the sample cases, was treated in much the same way as φ_B. Here we normalized each case to zero on the sheath side and switched sign on the magnetospheric side, if not already positive. This was to account for the facts that φ_V has no natural reference value in the magnetosphere and is determined more accurately in the magnetosheath.

Figures 8–10 show the results of the analysis. Figure 8 exhibits the ion and electron thermal properties whose changes defined the key times. It is thus by design that these four quantities have their changes centered at the key time. What is non-trivial though is the fact that the changes in thermal properties occur very abruptly, between neighbouring 10-s bins. In fact, inspection of the printouts of the data show that in many cases the change occurs inbetween subsequent measurements which are 4.4 s apart. As Figure 8 illustrates, the temperatures on average changed by a factor of 2 at the key time. Further inward, there are additional, more gradual changes which occur as the result of entry into the magnetosphere proper. These temperature changes are generally quite sudden and large in the individual cases, but are smeared out in the superposed epoch plots because, as a result of the varying boundary layer durations, they occur at different times relative to the key time.

Note that for the electrons the anisotropy parameter becomes positive (i.e., $T_{e\parallel} > T_{e\perp}$) at the key time, a fact which is true for most of the individual cases as well. For the ions $T_{\parallel}$ increases also at the key time, but

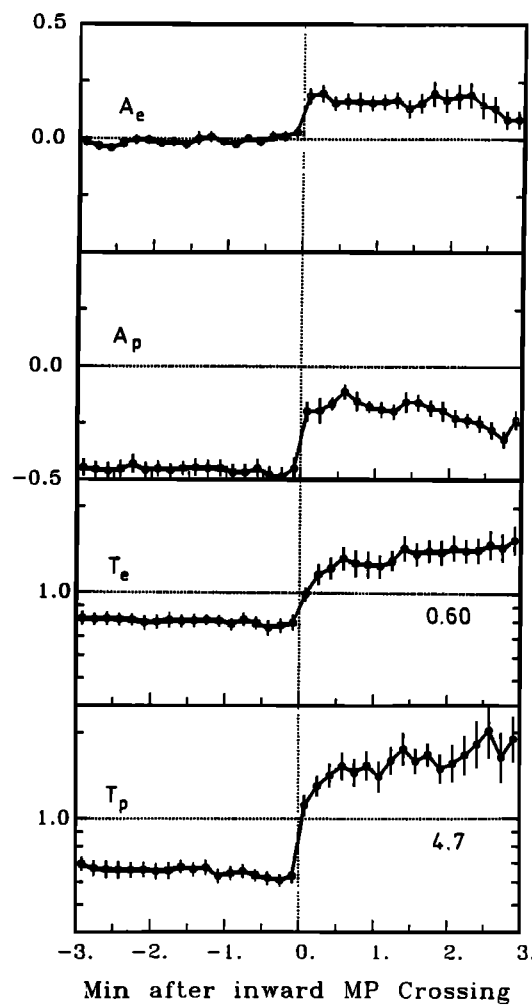

Fig. 8. Superposed epoch analysis of 20 low-shear magnetopause crossings. The key time (cf. Table 1) is defined by the time of the ion temperature change. The magnetosheath is on the left, the magnetosphere to the right of the key time. To comply with this definition, the time order of the outbound passes had to be reversed. From top to bottom, the figure displays the electron and ion temperature anisotropies, A_e and A_p, and the temperatures themselves, T_e and T_p (in 10^6K). The normalization of the averages is indicated by the numbers on the right. The vertical bars indicate the error of the mean. For details of the procedure see the text.

21562202a, 1993, A8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/93JA00646 by University College London UCL Library Services, Wiley Online Library on [10/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

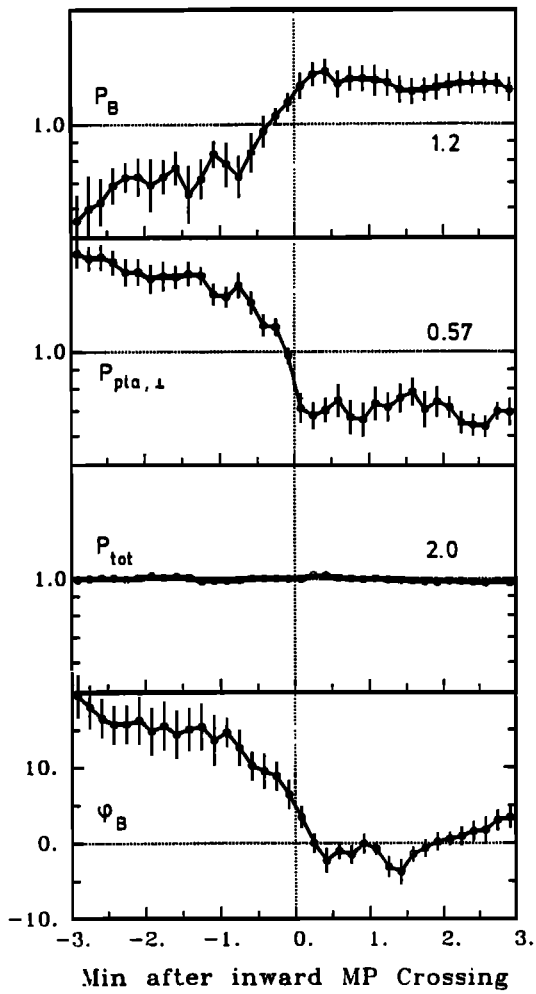

Fig. 9. Superposed epoch plots for the magnetic field pressure P_B, the perpendicular plasma pressure $P_{pla,\perp}$, the total pressure P_{tot} (all in nPa), and the magnetic field rotation angle φ_B.

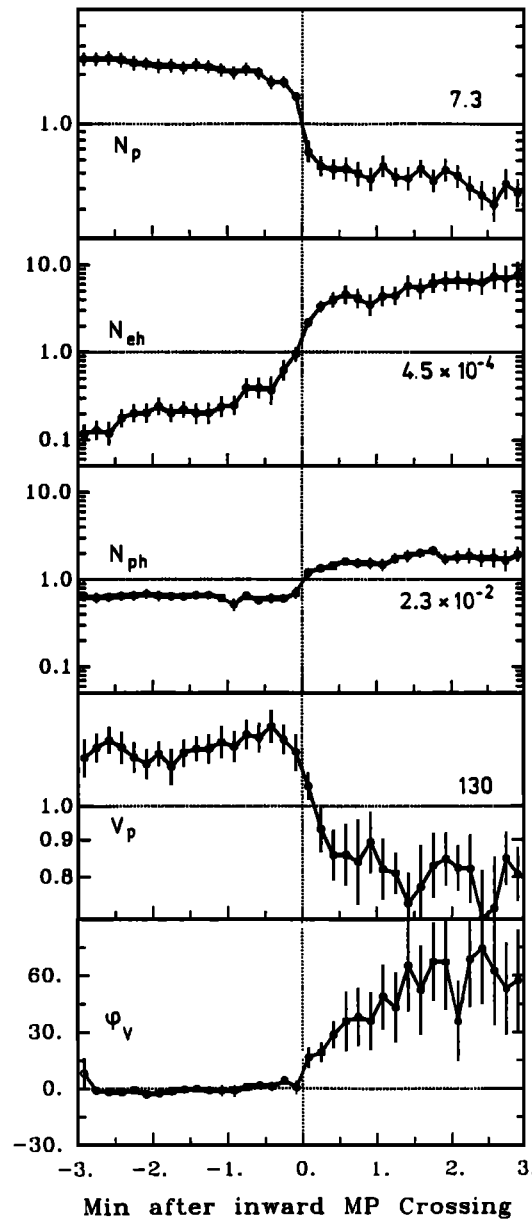

Fig. 10. Superposed epoch plots for the total plasma density, N_p, the partial densities of electrons between 1.8 keV and 30 keV and ions between 8 keV and 40 keV, N_{eh} and N_{ph} (all in cm^{-3}), and the magnitude, V_p (in km/s), and rotation angle, φ_V, of the plasma bulk velocity.

the anisotropy parameter remains negative (i.e., $T_{p\perp} > T_{p\parallel}$) throughout the interval.

Figure 9 shows the associated profiles of the plasma, magnetic, and total pressures, and the rotation angle of the magnetic field. Figure 10, finally, displays the profiles in total plasma number density, partial number densities of energetic electrons and ions, and plasma bulk speed.

Considering the pressures first, it is quite apparent that both the magnetic pressure as well as the (perpendicular) plasma pressure change significantly across the interval shown, but that the sum of the two (total pressure) remains remarkably constant. The constancy of the total pressure is required by the balance of normal stresses at the magnetospheric boundary and thus constitutes just a check on the validity and satisfactory absolute calibration of the data. The slow rise in magnetic pressure and drop in plasma pressure prior to the key time reflect the presence of a field pile up and plasma depletion region which we observe in about half the cases.

For the average plasma β (not shown), i.e., the ratio between plasma and magnetic field pressures, one obtains about 1.1 and 0.3 for the magnetosheath and magnetosphere sides, respectively.

The maximum in the magnetic field right at the key time or throughout the entire boundary layer, which we had noted in some of the individual cases presented earlier, is still evident in the magnetic pressure trace even though it is somewhat washed out by the averaging process due to the varying width of the peak.

Turning to the field direction (lowest panel in Figure 9), it is clear that on average there is about a 15° rotation of the field. Note that the individual crossings were selected to have field rotations less than about 30° but there were cases which had no noticeable rotation at all. Whatever field rotation is occurring seems to occur on both sides of the key time. Inspection of all individual crossings, however, has revealed that this is not always true. The field rotation can be limited to one side or the other of the key time. Thus the super-

21562202a, 1993, A8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/93JA00646 by University College London UCL Library Services, Wiley Online Library on [10/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

posed epoch analysis does not represent the timing of the field rotation well.

As the temperatures remain fairly constant on the sheath side (Figure 8), the gradual drop in plasma pressure implies a drop in density. This density depletion is clearly visible in Figure 10, and was already noted in the discussion of the individual cases. The density shows another, sharp drop at the key time, consistent with the fact that the temperature sharply rises at the key time while the pressure remains constant.

The partial densities of energetic electrons and ions increase around the key time, similar to what happens to the temperatures. This relationship is discussed in section 4.

The bulk speed first rises slowly, but then shows a large but fairly gradual drop across the magnetopause. The flow direction φ_V (bottom panel in Figure 10) is steady right up to the key time where it exhibits a sudden net change of about 15°. Thus the temperature boundary which defines the key time seems to also mark a flow boundary.

4. PLASMA DISTRIBUTION FUNCTIONS

We noted earlier that the rise in temperatures at the key time is usually accompanied by a rise in the partial densities of the energetic particles of magnetospheric origin. The question then is: Is the temperature increase simply caused by the admixture of another, more energetic population, or does it reflect true bulk heating? To address that question, we consider the measured electron and proton distribution functions. Those also shed light on the observed temperature anisotropies.

In Figure 11 we present a sequence of electron distribution functions from the September 17, 1984, crossing discussed earlier (Figure 2). The dashed and solid lines show the distribution functions parallel and perpendicular to the magnetic field, respectively, in terms of the logarithm of the phase space density versus velocity. The sequence begins in the magnetosheath (*a*) and progresses through the boundary layer (*b* and *c*) into the magnetosphere proper (*d*). The times where these distributions were taken have been marked in Figure 2 by vertical ticks along the electron temperature trace.

Comparison of distributions *a* and *b*, taken right before and after the magnetopause crossing, shows that the boundary layer distribution (*b*) is wider and less steep at the low velocities where the bulk of the plasma resides, and thus has a higher temperature. This is true for both the perpendicular and the parallel distributions. There also is an extra energetic component (of magnetospheric origin) above 20,000 km/s (~1 keV). Deeper in the boundary layer (*c*), the energetic component increases. Curve *d*, taken right after crossing the inner edge of the boundary layer, shows a distribution consisting of the energetic component, already noted in the boundary layer, and a component at low energies (up to 150 eV) which presumably is of ionospheric origin.

To distinguish true heating from mixing of different components, it is helpful to consider absolute phase space densities [cf. *Song et al.*, 1990*a*]. Figure 11 shows that at the lowest velocities curve *b* has lower values than curve *a*, reflecting the drop in plasma density which is observed at the time of the magnetopause crossing. (cf. Figure 2). At velocities between about 7500 and 15,000 km/s (160 - 650 eV), however, *b* has higher phase space densities than the sum of *a* and *d*. Thus in that energy range the boundary layer distribution cannot simply be obtained by mixing of the magnetosheath and magnetospheric components, even if both got into the boundary layer with undiminished intensity.

Turning to the anisotropies, we compare the solid and dashed lines in Figure 11 which represent the distributions perpendicular and parallel to the magnetic field, respectively. In the sheath the two distributions are nearly identical, implying $T_{e\perp} \sim T_{e\parallel}$. In the boundary layer the parallel distribution is slightly wider than the perpendicular one at low velocities where the phase-space density is high. The excess parallel flux is most pronounced at velocities just below 20,000 km/s (i.e., 650 eV).

At velocities of the order of 20,000 km/s (~1 keV) the parallel distribution drops below the perpendicular one (and below the 1-count level), i.e., the angular distribution becomes "trapped" at large velocities. This cross-over has little effect on the temperature though,

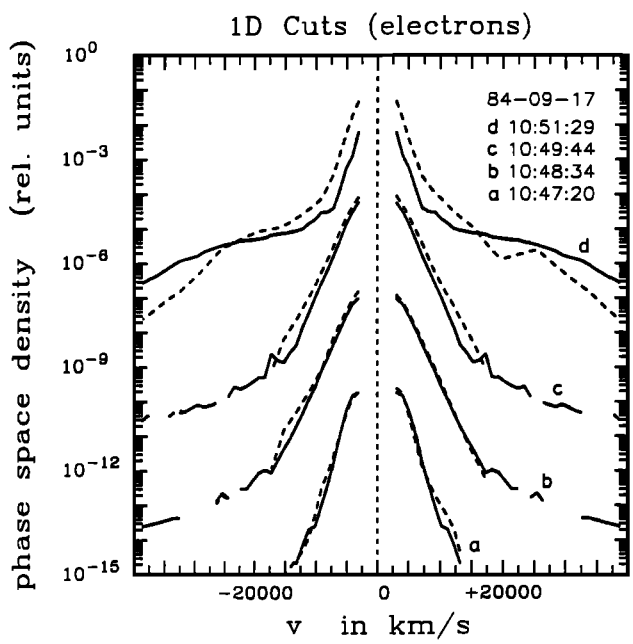

Fig. 11. Stacked electron distribution functions as measured on September 17, 1984, at the times indicated in the insert and marked in Figure 2. The bottom distribution (*a*) was measured in the magnetosheath, the next two (*b* and *c*) in the boundary layer, and the fourth (*d*) in the magnetosphere proper. Dashed lines represent cuts along the magnetic field through the interpolated three-dimensional distributions. The solid lines are obtained by averaging over all directions perpendicular to the field. For the latter, the right and left portions are redundant and shown as mirror images of each other. Distributions are displayed as log *f* versus velocity. The phase space density *f* is normalized to 10^{-25} s^3/cm^6, the velocity is in km/s. The scale applies to the uppermost curve, the other curves are displaced downwards by three decades each. The hole at center reflects the 15-eV low-energy cutoff of the electron measurements. The curves are interrupted or terminated whenever the underlying count rates fell below 1 count per 8.5-ms sample. The maximum speed shown corresponds to 4.6 keV.

21562202a, 1993, A8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/93JA00646 by University College London UCL Library Services, Wiley Online Library on [10/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

which is dominated by the low-energy fluxes, thus yielding $T_{e\parallel} > T_{e\perp}$.

In the magnetosphere proper (see curve *d* in Figure 11), the anisotropies and their cross-over become more pronounced. The strong excess in field-aligned flux at low energies is consistent with the component being of ionospheric origin. In contrast to the boundary layer cases the balance of low and high energy fluxes is now such that $T_{e\parallel} < T_{e\perp}$ overall.

Another sequence of electron distributions, for the August, 30, 1985, pass (cf. Figure 4), are shown in Figure 12. The same pattern as in Figure 11 is evident: the boundary layer distributions (*b*, *c*) are very similar in character to those observed in the adjacent magnetosheath (*a*), but because of their reduced slope and larger width by their smaller slope, have a distinctly higher temperature. (We note in passing that the magnetospheric distribution (curve *d*) again is composed of two distinct populations. This time the relative densities and temperatures of the two components are such as to mimic an overall effective temperature which is almost the same as that in the boundary layer, cf. Figure 4).

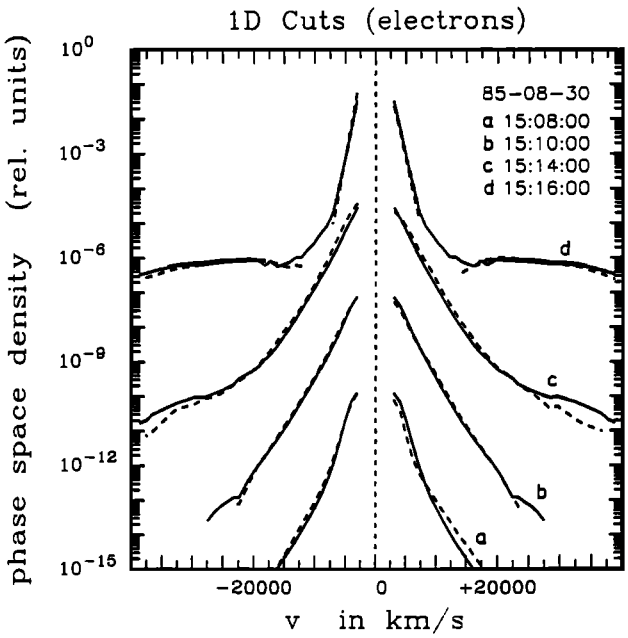

Fig. 12. Electron distributions for August 30, 1985 (cf. Figure 4). The format is the same as in Figure 11.

Electron distributions which demonstrate the occurrence of field-aligned anisotropies in the boundary layer have been reported earlier [*Ogilvie et al.*, 1984; *Hall et al.*, 1991]. Hall et al. also noted the transition from field-aligned to trapped angular distributions at energies of about 1 keV. *Thomsen et al.* [1987] have discussed electron and proton distributions for magnetospheric flux transfer events (FTEs) which are thought to mark sporadic encounters of the boundary layer. The field-aligned electron anisotropies are evident in those events also. Note that these papers dealt with mostly high-shear magnetopause crossings.

Ogilvie et al. and Hall et al. referred to the field-aligned electrons as "bidirectional beams" and "counterstreaming electrons", respectively. One of the interpretations they offer is that the electrons represent a separate (i.e., third) component injected or accelerated at a location along the field where the field magnitude was much stronger. Motion of these electrons along the field lines into the weaker field at the spacecraft location would then produce the noted field-aligned anisotropy. *Ogilvie et al.* [1984] cite as support for this interpretation the observations of such electrons at low altitudes on auroral field lines. More efficient injection or transport of magnetosheath electrons into the magnetosphere at high latitudes (stronger field) has been suggested by *Hall et al.* [1991], although high latitude does not automatically imply larger field strength. Excess parallel fluxes could also be the result of local heating occurring preferentially along the magnetic field, as noted by Hall et al.

Ogilvie et al. [1984] have concluded that the bi-directional streaming generally signifies closed field lines. *Hall et al.* [1991] have arrived at the same conclusion for the times the counterstreaming fluxes are equal in magnitude, but have taken unbalanced counterstreaming fluxes as indicating open field lines. Unbalanced counterstreaming electron fluxes, when considered in the framework of moments of the distribution functions, should generate non-zero parallel heat fluxes. We have examined the heat fluxes which we routinely compute but have in general found them too "noisy" to draw any firm conclusions. Heat fluxes are computed as integrals over the active energy range. Perhaps the restricted energy range (100–400 eV) which Hall et al. employ provides better sensitivity.

Proton distributions, for the same pass as in Figure 11, are shown in Figure 13. Here the distributions are taken through a point offset from the origin by the bulk

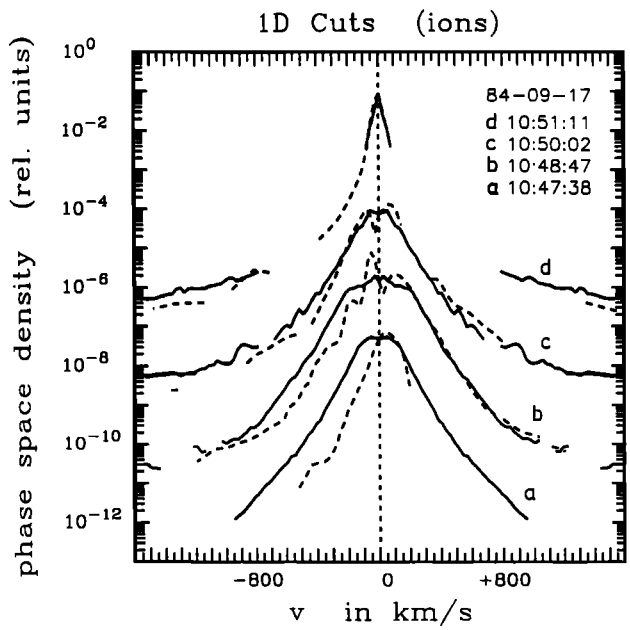

Fig. 13. Proton distributions for the same case as Figure 11. The parallel and perpendicular distributions (dashed and solid lines, respectively) are taken in the bulk velocity frame. The phase space density has been normalized to $10^{-20}\,s^3cm^{-6}$. The curves are displaced by 2 decades. The maximum velocity shown corresponds to 21 keV.

21562202a, 1993, A8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/93JA00646 by University College London UCL Library Services, Wiley Online Library on [10/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

velocity, and thus do not exhibit a hole at center. This offset introduces more interpolation and causes asymmetries in the count rates, the result of which are curves which are less smooth and which drop below sensitivity threshold at different velocities on the two sides.

Starting out in the magnetosheath (curve *a*), one notices the strongly different parallel and perpendicular distributions, consistent with the observed strong temperature anisotropy (cf. Figure 2). Going into the boundary layer, the parallel distribution approaches the perpendicular distribution. This way the anisotropy is reduced, the feature we have used to define the key time in our superposed epoch analysis. The admixture of energetic ions in the boundary layer is apparent, and so is the appearance of a low-energy component, presumably of ionospheric origin, in the magnetosphere proper (curve *d*). As already observed for the electrons, the magnetospheric energetic component cannot contribute significantly to the bulk of the boundary layer proton spectrum.

Returning now to the evidence for bulk heating, we note that for electrons we have established that the bulk of the boundary layer distribution cannot be explained as a mixture of magnetosheath and magnetospheric distributions. However, admixture of a suitably shaped third distribution could account for the excess fluxes. For the parallel direction, we have discussed evidence for such a component, presumably injected closer to the Earth. For the perpendicular direction, any third component would have to be created essentially locally. Heating of pre-existing cold ionospheric electrons would be a case at hand. It is however hard to believe that such a process would be able to account for an overall spectrum as smooth and featureless as exhibited by the bulk of the boundary layer electrons in Figures 11 and 12. We thus conclude that at least for the perpendicular direction it is likely that electrons are being genuinely heated upon entry into the boundary layer.

For protons the observed temperature increase is essentially in the parallel direction. Similar arguments for a third component as presented for the electrons then make it difficult to decide whether true heating occurs or not.

It should be noted that for some high-shear, near-tail magnetopause crossings, *Gosling et al.* [1986] have interpreted the elevated electron and proton temperatures in the boundary layer which they observe as being caused by bulk heating, and not by the admixture of more energetic particles. At least the latter then is consistent with the present finding.

5. DISCUSSION

5.1. *Magnetopause Definition and Identification*

The simplest definition considers the magnetopause as the region of current flow in the transition between the magnetosheath and the magnetosphere. In this paper we have been considering conditions of low magnetic shear across the magnetopause, and thus are dealing with cases where little or no current is needed to separate magnetosheath from magnetospheric field lines, at least locally. In this situation the above operational definition is not helpful. Worse, this definition would subsume under "magnetopause" the region we observe in about half the cases where current flows outside the magnetopause because the interplanetary magnetic field is piled up against the magnetospheric obstacle.

In this dilemma we turn to the observed changes in the plasma properties. If the magnetopause were an impenetrable boundary separating the solar wind from the magnetospheric plasma, the magnetopause could be easily identified, regardless of any magnetic field changes, because the two plasmas happen to have distinctly different densities and temperatures. The magnetopause is, however, not impenetrable, as evidenced by the usual presence of a boundary layer. Because of the boundary layer, it is not a priori clear what signatures to expect in the plasma properties which could be used to identify the magnetopause.

We have demonstrated in this paper that, in spite of the boundary layer, the magnetopause transition is not smooth but characterized by a distinct jump in plasma temperatures and temperature anisotropies, as well as an abrupt change in flow direction. It is tempting to attribute these changes, and in particular their often discontinuous nature, to the crossing of a boundary between topologically different magnetic fields, i.e., between closed magnetospheric and open interplanetary field lines.

There are, however, difficulties with that interpretation. As pointed out below, there is evidence that in 2 of the 22 cases there are open field lines Earthward of the temperature boundary, which would rule out the topological explanation at least for those two cases. There is also the fact, already noted earlier, that the small magnetic field rotations we observe have no fixed temporal relation to the key time where the changes in plasma properties occur. If a spatial interpretation applies, this would mean that the current flow associated with the field rotation has no fixed spatial relation to the plasma boundary.

Thus it appears that while the changes in plasma properties observed at the key time constitute a reliable signature of the magnetopause crossing for conditions of low magnetic shear, the interpretation of this plasma boundary, as well as the complete structure and width of the magnetopause, remain to be clarified.

5.2. *Magnetopause Thickness and Speed*

To translate the noted magnetopause time scales into spatial scales requires knowledge of the speed of the boundary relative to the spacecraft. With a single spacecraft that information is not readily available. One possibility is to use the measured instantaneous plasma velocity component along the magnetopause normal to deconvolve the parameter profiles and to determine the thickness, as done recently for the December 29, 1984, crossing [*Paschmann et al.*, 1990].

Taking a simpler approach, we have computed the average measured plasma velocity along model magnetopause normals for a 40 s interval centered on the magnetopause times for each of the 22 passes included in this study. The normal speeds thus obtained ranged from 3 to 71 km/s, with an arithmetic mean of 15 km/s and a standard deviation of 14 km/s. The geometric mean is 11 km/s. Assuming no bulk flow across the

21562202a, 1993, A8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/93JA00646 by University College London UCL Library Services, Wiley Online Library on [10/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

magnetopause, which is correct for a tangential discontinuity, the computed normal velocities represent the motion of the magnetopause relative to the satellite. In 3 of the 22 cases the sign of the average normal velocity differed from the sign inferred from the sense of the crossing (inbound versus outbound). Noting that we employed but model magnetopause normals this is probably not surprising.

For comparison, the timing analysis of a set of ISEE 1/2 crossings containing all shears [*Berchem and Russell*, 1982] has provided an average magnetopause speed of 40 km/s. No distinction according to magnetic shear was made in that study. Supporting evidence for smaller speeds for low-shear cases can be inferred, however, from another ISEE study [*Song et al.*, 1988] where it was found that the amplitude of magnetopause motion is less for northward than for southward IMF. If the periods of the oscillations were nearly identical for the two field directions, then the velocity must be smaller for the northward case.

As noted earlier, the temperature transition is often less than 4 s wide. Using our average magnetopause speeds, 4 s translate into 60 km or less. For comparison, a proton of average thermal speed in the average magnetopause magnetic field has a gyro radius of 60 km. As noted above, the temperature transition is perhaps embedded in a wider magnetopause structure, whose duration and width we have not been able to determine.

5.3. *Depletion Region*

Accepting the dashed line in Figures 8–10 as marking the outer edge of the magnetopause, the gradual drop in density preceding this time constitutes a plasma depletion layer. Figure 10 demonstrates the existence of a depletion layer only on average. The superposed epoch technique also smears out the duration of the feature. Inspection of the density profiles of all passes reveals that in 10 of the 22 cases a monotonic decrease in density, by a factor of two or more, is observed. Using less rigorous selection criteria would probably increase the percentage of depletion layer cases.

A plasma depletion layer was originally predicted by *Midgley and Davis* [1963] and *Lees* [1964], and treated in detail by *Zwan and Wolf* [1976]. Observational evidence for the occasional occurrence of a depletion layer was reported by *Paschmann et al.* [1978], *Crooker et al.* [1979], *Song et al.* [1990*b*], and *Fuselier et al.* [1991]. *Hall et al.* [1990, 1991] also consider plasma depletion but in a different framework, as discussed below.

The low-shear magnetopause crossing discussed earlier by *Song et al.* [1990*a*] may have contained a depletion layer, although these authors interpreted the broad region of density drop and field increase they observe as part of the magnetopause itself and called it the "sheath transition layer". Their Figure 1*a* shows that at the Earthward edge of this layer the ion temperature increases slightly, perhaps a sign that that is where according to our definition the magnetopause was crossed.

One of the predicted signatures of the plasma depletion layer is a drop in temperature anisotropy, i.e., a gain in perpendicular over parallel temperature, resulting from the combined effects of field compression and plasma depletion [*Crooker et al.*, 1979]. If the magnetic moment is preserved, the rise in field strength will be accompanied by a rise in $T_\perp$. Plasma depletion will preferentially remove particles with the largest parallel velocities, thus reducing $T_\parallel$. Decreases in $T_{p\parallel}/T_{p\perp}$ in the plasma depletion layer have been reported by *Fuselier et al.* [1991] and by *Anderson et al.* [1991].

Inspection of the individual depletion layer crossings reveals that in 7 of the 10 depletion layer cases the proton temperature anisotropy parameter becomes more negative, as expected. Examples are the cases displayed in Figures 2, 3, and 4. In three cases, however, the anisotropy parameter actually increases, as seen in Figure 7. Inspection of the individual traces for $T_{p\perp}$ and $T_{p\parallel}$ (not shown in those figures) reveals that it is a drop in $T_{p\parallel}$ which causes the anisotropy to change, while $T_{p\perp}$ stays almost constant. The latter implies that the magnetic moment is actually not conserved.

The duration of the depletion layer varies between 1 and 8 min, with an average over the 10 cases of approximately 3 min. Using the the magnetopause normal velocities inferred earlier (section 5.2), we have computed the thicknesses of the 10 individual cases. They ranged between 0.12 and 0.96 R_E, with an average of 0.40 R_E and a standard deviation of 0.28 R_E. The geometric mean is 0.31 R_E, or 2000 km. The model calculations of *Zwan and Wolf* [1976] give thicknesses between about 500 and 3000 km. Recent three-dimensional MHD calculations by *Wu* [1992] indicate depletion layer thicknesses of the order of 1 R_E. Our observations are thus in the expected range.

5.4. *Magnetic Field Overshoot*

In some of the individual crossings, and still visible in the superposed epoch results, overshoots in magnetic field strength right at the magnetopause, or over the entire boundary layer were observed. The appearance of such maxima just inside the magnetopause was noted long ago in OGO 5 data [*Neugebauer et al.*, 1974] and stressed again recently in the papers by *Hall et al.* [1990, 1991] which are based on AMPTE/UKS data.

Hall et al. [1990, 1991] observe a field maximum and concurrent minimum in electron pressure Earthward of the magnetopause in a number of cases. They suggest that the field compression and plasma depletion is generated when the field lines are draped around the region of closed field lines, rather than around the magnetopause (which they locate further outward). In a recent paper, *Sonnerup et al.* [1992] have demonstrated that the maximum in B and concurrent minimum in electron (and ion) pressures can be accounted for by mechanisms not invoking field pile-up and plasma depletion, having to do with field curvature and/or energetic particle pressure.

5.5. *Boundary Layer*

From the survey of the individual crossings it has been determined that the duration of the boundary layer encounters varied strongly, from about 4 s to about 14 min. The character of the profiles ranges from smooth gradual decays to a sequence of pulses. There are a number of cases where there is a sharp density drop at the magnetopause, followed by a plateau, and another sharp drop at the transition to the magneto-

21562202a, 1993, A8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/93JA00646 by University College London UCL Library Services, Wiley Online Library on [10/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

sphere proper. Strong spatial and temporal variations are known to occur in the boundary layer, but with a single spacecraft they cannot be untangled. No dependence on local time was apparent.

Averaging the measured plasma velocities along the magnetopause normal over the individual boundary layer passes, we obtained thicknesses ranging from 0.02 to 1.1 R_E, with an average of 0.30 R_E and a standard deviation of 0.30 R_E. The geometric mean is 0.17 R_E, or 1100 km. Note that in only 3 of the 22 cases the sign of the average normal velocity was inconsistent with the sense of the crossing. In two of these latter cases the normal speed was quite low (less than 6 km/s) and in the other it was extremely variable. One should also remember that we employed model magnetopause normals and kept them fixed for the entire boundary layer crossing.

The inferred average normal speed in the boundary layer was 11 km/s, very similar to what we had obtained for the magnetopause itself. As noted earlier, there is reason to believe that the boundary speeds are systematically higher for large shear situations. Thus one has to be careful with conclusions about systematic differences in boundary layer thickness for low and high shear, based solely on observations of their duration. We also find that there is a tendency for anticorrelated boundary layer durations and speeds. As a result, the inferred boundary layer thicknesses vary much less than their durations.

As mentioned earlier, the boundary layer can be distinguished in almost all cases from the magnetosheath by the electron temperature anisotropy. While for the sheath the anisotropy is smaller than or equal to zero, in the boundary layer it becomes nearly always positive (i.e., $T_\parallel > T_\perp$), 0.2 on average. This feature greatly helps to identify the boundary layer. By contrast, the ion temperature anisotropy, even though it usually changes by a larger amount than that of the electrons, remains less than zero on either side of the boundary, and therefore is of much less help in differentiating the regions.

The behavior of the electron temperature anisotropy is in agreement with the results from AMPTE/UKS [*Hall et al.*, 1991]. It is also consistent with the electron angular distributions reported by *Ogilvie et al.* [1984] and by *Thomsen et al.* [1987] which we have referred to in Section 4 and which were primarily dealing with high-shear magnetopauses. One therefore would conclude that the field-aligned electron anisotropies are a general property of the boundary layer.

Although the presence of such field-aligned anisotropies has generally been taken as evidence for closed magnetic field lines (cf. Section 4), it should be remembered that for the November 10, 1984, case *Hall et al.* [1990, 1991] have inferred the presence of open field lines in the outer portion of the boundary layer, based on the electron angular distributions they observe. For the November 10 and the December 29, 1984, cases we have shown in an earlier paper [*Paschmann et al.*, 1990] by a test of the Walén relation that the magnetic field locally had a nonzero, although probably very small, normal component. Thus it could be inferred that at least the outermost part of the boundary layer in these cases was located on open field lines.

5.6. *Magnetopause Processes*

For the conditions we have studied in this paper, where the magnetic fields on the two sides of the low-latitude magnetopause are nearly parallel, one expects magnetic reconnection in the subsolar region not to play a role. Thus the low-shear data set ought to be particularly useful for studying non-reconnection transfer processes. One would then tend to take the fact that we are observing a boundary layer in all crossings as proof that non-reconnection transfer processes are indeed occurring, and that these processes are acting locally. However, neither conclusion is safe. First, there is no assurance that the plasma entered locally, it could equally well have entered somewhere else and been transported to the our observation site at a later time. Second, reconnection at a remote site, in particular the polar cusps, could provide boundary layer plasma [e.g., *Cowley*, 1981; *Song and Russell*, 1992]. In fact, it has been suggested that the cases referred to above where we have found evidence for finite normal magnetic fields for two of our low-shear cases could be interpreted in exactly that way [*Paschmann et al.*, 1990].

As to the role of non-reconnection processes, important information should be contained in the parameter profiles across the boundary layer. For example, one would hope to infer the presence or absence of diffusion from the character of the density profiles. This would require knowledge of the true spatial profiles, which as noted above is not readily available.

An important finding of our study with regard to the plasma entry processes concerns the nature of the temperature changes observed at the magnetopause. In principle, these could be caused either by bulk heating of the plasma upon entry from the magnetosheath into the boundary layer, or as a result of the observed admixture of hot magnetospheric plasma. In section 4 we have demonstrated that the bulk plasma itself is hotter in the boundary layer. Unless one is willing to postulate that the entry is always occurring at a remote place where the magnetosheath plasma happens to be hotter than at our observation site, this finding means that the plasma is truly heated upon entry. By what mechanism remains to be explained.

6. SUMMARY

In this paper we have established a method, based on changes in plasma thermal properties, to identify the magnetopause under conditions of low ($\leq 30°$) magnetic shear. This temperature boundary occurs simultaneously with a plasma flow boundary. It is tempting to attribute both to the crossing of a topological boundary, but that interpretation seems to be inconsistent with independent deductions on field topology at least in two cases. It also appears that the narrow temperature transition is embedded in a wider magnetopause current structure.

In all 22 low-shear crossings examined a magnetopause boundary layer was observed, consistent with earlier inferences that solar wind plasma enters the magnetosphere regardless of magnetic field orientation. We find the same electron temperature anisotropy (i.e., $T_\parallel > T_\perp$) in the boundary layer that has been reported

21562202a, 1993, A8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/93JA00646 by University College London UCL Library Services, Wiley Online Library on [10/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

earlier for high-shear magnetopauses. On the magnetosheath side the ratio is usually ≤ 1. The electron temperature anisotropy is thus uniquely suited to distinguish these regions.

We have looked at the evidence for a layer of plasma depletion magnetic field pile-up on the magnetosheath side. Requiring that the density drops monotonically and by at least a factor of two, we have identified such a layer in 10 of the 22 cases. In the majority of the 10 cases, we observe a drop in $T_{p\|}$ as expected in models of density depletion.

Analysis of the electron and proton distribution function reveals that the temperature increase which we identify with the magnetopause crossing is provided by the bulk plasma, and not by the admixture of the more energetic magnetospheric particles. This implies that entry into the boundary layer is associated with heating.

Using the component of the measured plasma bulk velocities along the magnetopause normal direction, we have inferred the thickness and speed of the magnetopause and its surrounding layers. For the average speed we obtain 11 km/s, an indication that the magnetopause moves more slowly and/or with reduced amplitude for low-shear conditions. Such a shear-dependent speed implies that one should be careful drawing conclusions on boundary layer thickness as a function of IMF direction which are based simply on observed crossing durations.

The temperature transition can be as thin as 50 km or less, comparable with typical proton gyro radii at the magnetopause. For the boundary layer and the plasma depletion layers, we obtain average thicknesses of $0.3\,R_E$ and $0.4\,R_E$, respectively.

Acknowledgments. The authors thank B. U. Ö. Sonnerup for helpful discussions and comments on an earlier version of the manuscript. The constructive criticisms and helpful suggestions of both referees are gratefully acknowledged.

The Editor thanks P. Song and N. Crooker for assistance in evaluating this paper.

REFERENCES

Anderson, B. J., S. A. Fuselier, and D. Murr, Electromagnetic ion cyclotron waves observed in the plasma depletion layer, *Geophys. Res. Lett.*, *18*, 1955–1958, 1991.

Berchem, J., and C. T. Russell, The thickness of the magnetopause current layer: ISEE 1 and 2 observations, *J. Geophys. Res.*, *87*, 2108–2114, 1982.

Cowley, S. W. H., Magnetospheric and ionospheric flow and the interplanetary magnetic field, in *The Physical Basis of the Ionosphere in the Solar-Terrestrial System*, p. 4-1, NATO, Neuilly-sur-Seine, 1981.

Crooker, N. U., T. E. Eastman, and G. S. Stiles, Observations of plasma depletion in the magnetosheath at the dayside magnetopause, *J. Geophys. Res.*, *84*, 869–974, 1979.

Fuselier, S. A., D. M. Klumpar, E. G. Shelley, B. J. Anderson, and A. J. Coates, He^{2+} and H^+ dynamics in the subsolar magnetosheath and plasma depletion layer, *J. Geophys. Res.*, *96*, 21,095–21,104, 1991.

Gosling, J. T., M. F. Thomsen, S. J. Bame, and C. T. Russell, Accelerated plasma flows at the near-tail magnetopause, *J. Geophys. Res.*, *91*, 3029–3041, 1986.

Haerendel, G., G. Paschmann, N. Sckopke, H. Rosenbauer, and P. C. Hedgecock, The frontside boundary layer of the magnetosphere and the problem of reconnection, *J. Geophys. Res.*, *83*, 3195–3216, 1978.

Hall, D. S., M. A. Hapgood, and D. A. Bryant, The transition from the magnetosheath to the magnetosphere, *Geophys. Res. Lett.*, *17*, 2039–2042, 1990.

Hall, D. S., C. P. Chaloner, D. A. Bryant, D. A. Lepine, and V. P. Tritakis, Electrons in the boundary layers near the dayside magnetopause, *J. Geophys. Res.*, *96*, 7869–7891, 1991.

Lees, L., Interaction between the solar plasma wind and the geomagnetic cavity, *AIAA J.*, *2*, 1576–1582, 1964.

Lühr, H., N. Klöcker, W. Oelschlägel, B. Häusler, and M. Acuña, The IRM fluxgate magnetometer, *IEEE Trans. Geosci. Remote Sens.*, *GE-23*, 259–261, 1985.

Midgley, J. E., and L. Davis, Calculation by a moment technique of the perturbation of the geomagnetic field by the solar wind, *J. Geophys. Res.*, *68*, 5111–5123, 1963.

Neugebauer, M., C. T. Russell, and E. J. Smith, Observations of the internal structure of the magnetopause, *J. Geophys. Res.*, *79*, 499–510, 1974.

Ogilvie, K. W., R. J. Fitzenreiter, and J. D. Scudder, Observations of electron beams in the low latitude boundary layer, *J. Geophys. Res.*, *89*, 10,723–10,732, 1984.

Paschmann, G., N. Sckopke, G. Haerendel, I. Papamastorakis, S. J. Bame, J. R. Asbridge, J. T. Gosling, E. W. Hones, Jr., and E. R. Tech, ISEE plasma observations near the subsolar magnetopause, *Space Sci. Rev.*, *22*, 717–737, 1978.

Paschmann, G., H. Loidl, P. Obermayer, M. Ertl, R. Laborenz, N. Sckopke, W. Baumjohann, C. W. Carlson, and D. W. Curtis, The plasma instrument for AMPTE/IRM, *IEEE Trans. Geosci. Remote Sens.*, *GE-23*, 262–266, 1985.

Paschmann, G., B. Sonnerup, I. Papamastorakis, W. Baumjohann, N. Sckopke, and H. Lühr, The magnetopause and boundary layer for small magnetic shear: Convection electric fields and reconnection, *Geophys. Res. Lett.*, *17*, 1829–1832, 1990.

Russell, C. T., and R. C. Elphic, Initial ISEE magnetometer results: Magnetopause observations, *Space Sci. Rev.*, *22*, 681–715, 1978.

Song, P., and C. T. Russell, Model of the formation of the low-latitude boundary layer for strongly northward interplanetary magnetic field, *J. Geophys. Res.*, *97*, 1411–1420, 1992.

Song, P., R. C. Elphic, and C. T. Russell, ISEE 1 & 2 observations of the oscillating magnetopause, *Geophys. Res. Lett.*, *15*, 744–747, 1988.

Song, P., R. C. Elphic, C. T. Russell, J. T. Gosling, and C. A. Cattell, Structure and properties of the subsolar magnetopause for northward IMF: ISEE observations, *J. Geophys. Res.*, *95*, 6375–6387, 1990a.

Song, P., C. T. Russell, J. T. Gosling, M. Thomsen, and R. C. Elphic, Observations of the density profile in the magnetosheath near the stagnation streamline, *Geophys. Res. Lett.*, *17*, 2035–2038, 1990b.

Sonnerup, B. U. Ö., G. Paschmann, T.-D. Phan, and H. Lühr, Magnetic field maxima in the low latitude boundary layer, *Geophys. Res. Lett.*, *19*, 1727–1730, 1992.

Thomsen, M. F., J. A. Stansberry, S. J. Bame, S. A. Fuselier, and J. T. Gosling, Ion and electron velocity distributions within flux transfer events, *J. Geophys. Res.*, *92*, 12,127–12,136, 1987.

Wu, C. C., MHD flow past an obstacle: Large-scale flow in the magnetosheath, *Geophys. Res. Lett.*, *19*, 87–90, 1992.

Zwan, B. J., and R. A. Wolf, Depletion of solar wind plasma near a planetary boundary, *J. Geophys. Res.*, *81*, 1636–1648, 1976.

W. Baumjohann, G. Paschmann, T.-D. Phan, and N. Sckopke, Max-Planck-Institut für extraterrestrische Physik, D-8046 Garching, Germany.

H. Lühr, Institut für Geophysik und Meteorologie, Technische Universität, Mendelssohnstr. 3A, D-3300 Braunschweig, Germany.

(Received October 6, 1992;
revised February 4, 1993;
accepted March 8, 1993.)

21562202a, 1993, A8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/93JA00646 by University College London UCL Library Services, Wiley Online Library on [10/12/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License